PREP 4 · LESSONS 7–8

Your investigation

Planning a fair test · dealing with odd repeats · evaluating and improving a method

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Explain why control variables must be kept the same. DN 873 [1]

If two variables change together, you cannot tell which one caused the result.

2 State how uncertainty is calculated from a set of repeated readings. DN 856 [1]

Uncertainty = (largest reading − smallest reading) ÷ 2

3 What is an anomalous result? DN 895 [1]

A result that does not fit the pattern of the others.

Key ideas LESSONS 7–8

Show your working

4 A group investigates how the **length of a pendulum** affects the **time for 10 swings**.

a Name two control variables. [1]

Any two: mass of the bob, starting angle (how far it is pulled back), the same person timing ✓

b Explain why they time 10 swings instead of one. [1]

One swing is very short; the reaction-time error is a much smaller part of a longer time (so the result is more accurate) ✓

5 One row of their results:

Length / cm	Trial 1 / s	Trial 2 / s	Trial 3 / s	Mean / s
40	12.6	12.8	15.9	12.7

a Which trial is anomalous? Trial 1 | Trial 2 | **Trial 3** [1]

b Calculate the mean, leaving out the anomalous result. Write it in the table. [1]

(12.6 + 12.8) ÷ 2 = 25.4 ÷ 2 = 12.7 s ✓

6 Evaluate the method: give **one** improvement, and explain how it makes the results better. [2]

Improvement, e.g. use a fiducial (marker) line at the centre of the swing / use a light gate / take more repeats ✓

Why, e.g. easier to see when to start and stop the watch, so less random error ✓

7 Their graph is a curve. Should they join the points dot-to-dot? yes | **no** Give the reason. [1]

No: draw one smooth best-fit curve showing the trend, not every reading ✓